Spot Call



Long call.

long S@K.

payoff = $S_T - K$.

F_T

T_1 ≤ T_2 = 1500

Long call futures.

long F@K = 1200

Close out@F

1400 - 1200 = 200. = payoff = $F_{1e} - K$

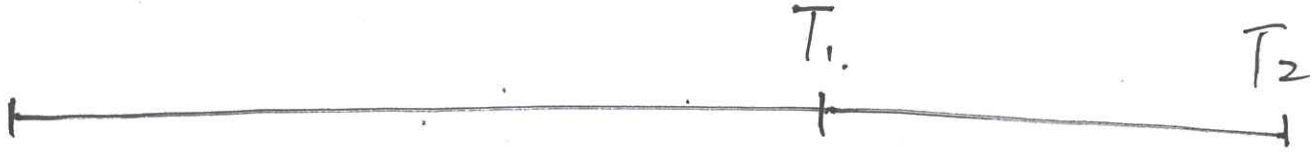
Payoff = $F_2 - F_1$

1400

Payoff = $F_2 - K$.

Payoff = $F_2 - K$ = 1500 - 1200 = 300

1500 - 1400 = 100



Long put futures.

Short F@K.

payoff = $K - F_1$

Close out F_2

Payoff = $F_1 - F_2$

Payoff = $K - F_2$.

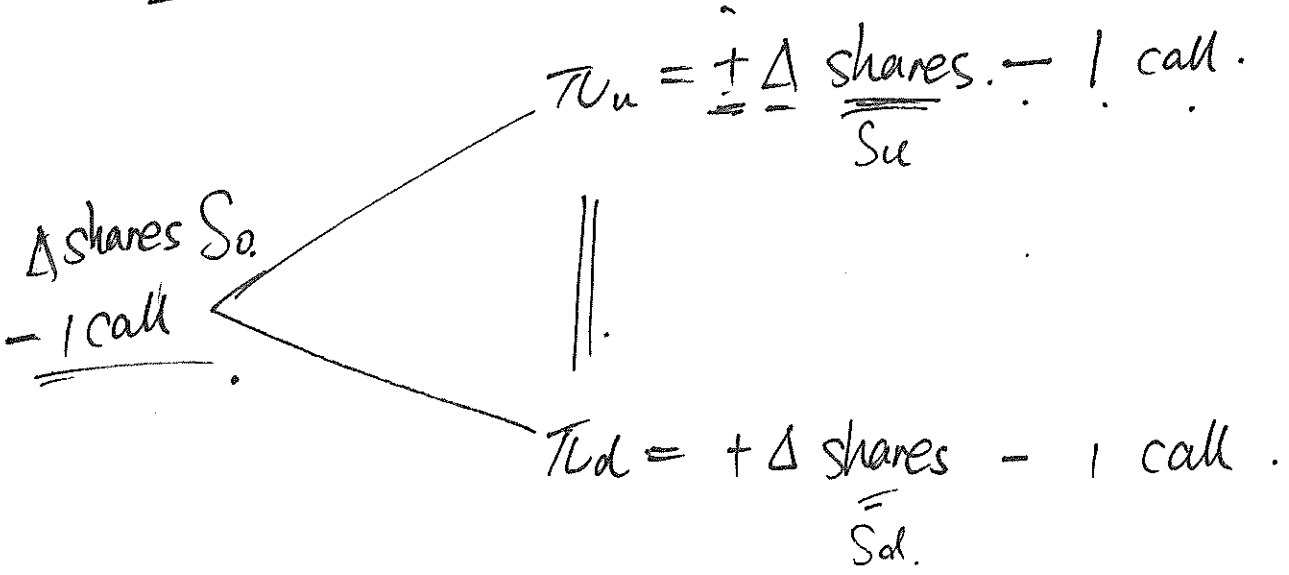
	0	T	
		$F_T \geq K$	$F_T < K$
A.	c $K \cdot e^{-rT}$	$F_T - K$ K	0 K
	$c + K \cdot e^{-rT}$	F_T	K
B. cash	P $F_0 \cdot e^{-rT}$	0 F_0	$K - F_T$ F_0
	0 (long F)	$F_T - F_0$	$F_T - F_0$
	$P + F_0 \cdot e^{-rT}$	F_T	K

$$c + K \cdot e^{-rT} = P + F_0 \cdot e^{-rT}$$

$$c + K \cdot e^{-rT} = P + S_0$$

$$S_0 = F_0 \cdot e^{-rT} \qquad F_0 = S_0 \cdot e^{rT}$$

Spot Call. long. short.



$$\pi_u = \pi_d = \Delta \cdot S_u - f_u = \Delta \cdot S_d - f_d.$$

$$\Delta \cdot S_0 - f = PV(\pi_u).$$

long short.

$C < F_0 - K$

$C + K < F_0$

long short.

long C., imm. exercise, long F @ K.,

short F_0.

$F_0 - (K + C) > 0.$

Call. $K = \$29$.

long $\Delta \cdot F_0 = \$30$.
- 1 call.
f.

Π_0

$F_u = 33$. $f_u = 33 - 29 = 4$.

$\Pi_u = (33 - 30) \cdot \Delta - 4$

$F_d = 28$. $f_d = 0$.

$\Pi_d = (28 - 30) \cdot \Delta - 0$

Value of Futures.

long $F = F_t - F_0$.

$\therefore$

$$(33 - 30)\Delta - 4 = (28 - 30)\Delta - 0$$

$$3\Delta - 4 = -2\Delta$$

$$\Delta = 0.8.$$

$$\Pi_0 = \Delta \cdot 0 - f = -1.592.$$

$$-f = -1.592.$$

$$f = 1.592.$$